



Comparative binding studies of curcumin and tangeretin on up-stream elements of NF- κ B cascade: a combined molecular docking approach

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Abstract

The hyper-activation of NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells) in cellular system is associated with inflammation, cancer and other human diseases. The core elements of NF- κ B cascade might regulate the translocation and activation of NF- κ B transcription factor. In the present study, we focused on active site analysis and molecular docking studies of core elements of NF- κ B pathway. Curcumin and tangeretin are the natural compounds used as ligands in this docking study. We performed comparative docking studies by applying individual and combination of ligands on drug target proteins (NF- κ B core elements). The docking energy and active site residues interacts with curcumin and tangeretin were obtained as follows: IKK complex (-4.13 kcal/mol; Leu⁷³⁷, Asp⁷³⁸, Trp⁷³⁹, Lys⁹⁶ and -2.76 kcal/mol; Thr⁷³⁵, Ser⁷³³), proteasome (-4.38 kcal/mol; Gln⁴⁹, Asp⁵², Arg⁴², Glu²⁴ and -4.05 kcal/mol; Asp⁵²), RelA (-6.00 kcal/mol; Arg³⁰⁴, Thr³⁰⁸, Arg³⁰² and -3.46 kcal/mol; Arg³⁰², Thr³⁰⁵) and NF- κ B (-5.42 kcal/mol; Cys³⁵⁹, Lys⁴⁴⁵, Lys⁵⁴¹, Glu³⁶⁰ and -5.31 kcal/mol; Lys⁴⁴⁴, Tyr³⁵⁷), respectively. Hex combined docking score obtained for IKK complex; proteasome, RelA and NF- κ B are -88.28 , -98.64 , -53.89 and -80.55 kcal/mol, respectively. In this study, we found that proteasome has shown the highest docking energy (-98.64 kcal/mol) and it is suggested to be as crucial drug target in NF- κ B cascade. However, further experimental and clinical verification is needed for clear evidence.

Keywords NF- κ B · Cancer · Inflammation · Curcumin · Tangeretin · Proteasome · RelA · HEX docking

1 Introduction

Signaling pathways are less controlled in carcinogenesis. Abnormal regulation of signaling pathways play a key role in cancer and other inflammatory diseases (Chen et al. 2006). NF- κ B signaling pathway is a complex network linking extracellular stimuli to cell survival and proliferation. It is activated in response to variety of activators such as viruses, cytokines, interleukins, lipopolysaccharides (LPS), oxidative

stress, and DNA-damaging agents (Baldwin 1996; Baeurle and Baltimore 1996; Garg and Aggarwal 2002). NF- κ B is implicated in inflammation, immune responses and origin of various solid tumors (Ghosh et al. 1998). On the other hand, NF- κ B signaling is a key to immune function and is likely necessary for antitumor immunity. Our present study is to challenges the designing of therapeutic approaches for targeting NF- κ B signaling pathway.

NF- κ B activates in classical or alternative signaling pathways (Sethi et al. 2008). These pathways are stimulated by different signals and require separate I κ B kinase (IKK) complexes. Many receptors activate predominantly the alternative pathway. A substantial therapeutic target development has been focused on the classical signaling pathway (Baldwin 2001).

The active form of NF- κ B consists of homo and hetero dimers of the NF- κ B/Rel family proteins [p50/p105 (NF- κ B1), p52/p100 (NF- κ B2), c-Rel, RelB, and p65 (RelA)]. In mammalian cells, NF- κ B exists in inactive form in the

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cytosol and which is associated with inhibitory I κ B proteins (I κ B α , I κ B β , I κ B γ). The NF- κ B activation in the cytosol requires sequential phosphorylation, polyubiquitination- and proteasome-mediated degradation of I κ Bs, and subsequent dissociation of the NF κ B-I κ B complexes (Baeurle and Henkel 1994; Verma et al. 1995; Baeurle and Baltimore 1996). The active form of NF- κ B complex translocate into nucleus thereby initiate the transcription of genes related to cytokines, chemokines, cyclins and anti-apoptotic proteins (Karin 2006).

IKK Kinase complex is the first core element of NF- κ B signaling pathway. It is essentially made up of two kinases (IKK α and IKK β) and one regulatory subunit, NEMO/IKK γ (Yamaoka et al. 1998; Israël 2010). Although IKK β phosphorylation is necessary for activation of the canonical pathway, whereas phosphorylation of IKK α is required for activation of the alternative pathway. Phosphorylation of I κ Bs marks them out for destruction, thereby relieving their inhibitory effect on NF- κ B. Subsequent release of NF- κ B translocates into the nucleus. Inhibition of IKK complex activation is one of the important ways to block the NF- κ B translocation. Carvalho et al. (2007) has been reported that a cell-permeable peptide that analogous to the leucine zipper subdomain of IKK γ , preventing NEMO oligomerisation, inhibited the constitutive NF- κ B activation and induced apoptotic cell death in human myelodysplastic syndromes (MDS) and acute myeloid leukemia (AML). NF- κ B inhibition by proteasome inhibitor (bortezomib) or IKK β inhibitor [kinase inhibitor of NF- κ B-1 (KINK-1)] significantly enhance the antitumor activity of camptothecin (Amschler et al. 2010).

The 26Sproteasome is the crucial site for intracellular protein degradation. The proteolytic core of 26Sproteasome complex consists of multiple peptidase activities, including the chymotrypsin-like (CT-like), trypsin-like (T-like), and peptidylglutamyl peptide hydrolyzing like. The unneeded or damaged proteins of cell cycle, DNA repair, apoptosis, translation, transcription and other cellular metabolisms are degraded by proteolysis through proteosomal pathway. This structure is found in the nucleus and the cytosol of all cells. In the NF- κ B cascade, proteasome mediates the I κ B degradation, thereby subsequent activation of NF- κ B transcription factor (Traenckner et al. 1994). The pharmacological approaches such as proteasome inhibitors and inhibitors of E1/E2 enzymes, E3 ubiquitin ligases are targeted to proteasome-ubiquitin system. Several studies have been shown that inhibition of CT-like activity in tumor cells is a strong stimulus that induces apoptosis. Milacic et al. (2008) showed that curcumin inhibits the CT-like activity of a purified rabbit proteasome (IC₅₀ = 1.85 μ M) and 26Sproteasome in human colon cancer cell lines. Bortezomib, the proteasome inhibitor has been currently used in the treatment of myeloma and lymphoma (Ludwig et al. 2005; Denz et al. 2006).

NF- κ B active complex consists of two subunits: NF- κ B p65 and p50, both of which are share sequence homology at N-terminal with the v-rel (reticuloendotheliosis viral oncogene homolog) products of oncogenes. The transactivation and I κ B binding property of NF- κ B is resided within the p65 subunit. The Rel homology domain of p65 protein has nuclear localization sequence (NLS), it consists of four highly conserved basic residues: Lys-Arg-Lys-Arg (KRKR). The NLS on p65 is masked by the I κ B in the cytosol and prevent the translocation of the NF- κ B (Urban and Baeurle 1975). Furthermore, NLS has been shown to be the target for receptor proteins that are involved in nuclear transport (Adam and Gerace 1991). Thus, one possibility is to prevent the interaction of NLS and nuclear transport proteins is the novel approach for the inhibition of NF- κ B mediated gene expression. Baicalein (plant flavonoid) and 3,5-bis(2-pyridinylmethylidene)-4-piperidone (EF31), a curcumin analog exhibited significantly greater inhibition of NF- κ B nuclear translocation (Seo et al. 2011; Olivera et al. 2012).

In another way, blockade of the interaction between NF- κ B and DNA in the nucleus may be supposed to inhibit the NF- κ B activation. Several natural compounds or synthetic agents have been shown inhibitory action on NF- κ B activation in cancer cells (Gupta et al. 2010; Yamamoto and Gaynor 2001).

Chemotherapy could not be fully effective in combatting high incidence of cancers. Further, it is accompanied by various side effects, toxicity and the development of chemotherapeutic resistance. Treatment of some cytostatic agents (camptothecin and other anthracycline drugs) causes chemoresistance in tumor malignancies (Ivanov et al. 2003). Combined chemotherapeutics approaches are increasing in cancer treatment to reduce the tumor resistance mechanisms. However, the molecular mode of action of inhibitors appeared to be conspicuously different. The combination therapy of chemotherapeutics agents are higher cost and causes adverse effects (Meyerhardt and Mayer 2005). To overcome the side effects caused by chemotherapeutic agents, administration of plant bioactive compounds that are nontoxic and capable of selectively arresting cell growth and inducing apoptosis in cancer cells (Al-Qubaisi et al. 2011). Thus, the present study is designed to investigate the effects of curcumin and tangeretin combination approach on up-stream elements of NF- κ B pathway. The combination effect of curcumin and tangeretin on NF- κ B inhibition was not studied earlier.

Curcumin is the major constituent of turmeric powder extracted from rhizomes of the *Curcuma longa* L. It shows strong antiproliferative, antioxidant, anticancer and anti-inflammatory activities (Kuo et al. 1996; Chen et al. 1999; Walters et al. 2008; Ravindran et al. 2009) in in vivo and in vitro. Curcumin inhibits TNF- α induced NF- κ B activation in human myelomonoblastic leukemia cells (Huang et al. 1991; Singh and Aggarwal 1995). The detailed molecular mechanism

of NF- κ B inhibition by curcumin was still unclear (Kumar et al. 1998). Curcumin can effectively inhibit the NF- κ B and HDAC signaling in liver cancer patients (Marquardt et al. 2015). Combination of curcumin with chemotherapeutic agents such as cisplatin, doxorubicin and dasatinib resulted in a synergistic anti-tumour activity (Notarbartolo et al. 2005; Nautiyal et al. 2011). Curcumin involves in the regulation of various signaling pathways in carcinogenesis (Choudhuri et al. 2002); therefore, cytotoxic therapies combined with this compound may exert enhanced anti-tumour effects through synergistic actions and may also decrease systemic toxicity. Curcumin and triptolide (a diterpenoid epoxide produced by the thunder god vine, *Tripterygium wilfordii*) could synergistically inhibit ovarian cancer cell growth, and induce apoptosis in ovarian cancer cell lines (Cai et al. 2013).

Tangeretin is an *O*-polymethoxylated flavone occurs in tangerine (*Citrus tangerine* L.), orange colored citrus fruits and other citrus peels. It induces apoptosis and cause cell cycle arrest through modulation of key G1 regulatory proteins such as cdk2, cdk4, and cdk inhibitors- p21 and p27 in cancer cells (Pan et al. 2002; Choi et al. 2007). It performed cytotoxic activity with IC₅₀ less than 5 μ M on melanoma cells (Yanez et al. 2004) and inhibits COLO 205 colon cancer cells through inhibiting retinoblastoma (Rb) phosphorylation and CDK2/4 inhibition mediated G1 arrest (Pan et al. 2002). Researchers exhibit potency of tangeretin as chemoprevention agent. Co-combination of tangeretin and nobiletin inhibited the histamine-induced activation of NF- κ B, AP-1, TNF- α , IL-4 and p38 in mice and PMA-stimulated RBL-2H3 cells (Jang et al. 2013). Tangeretin enhanced the tamoxifen's therapeutic effect in vivo. Combination of tamoxifen and tangeretin exhibit inhibitory effects on tumor growth and invasive properties of cancer (Bracke et al. 1999). Meiyanto et al. (2011) showed tangeretin improve the anticancer effect of doxorubicin on MCF-7 and T47D cells by inducing apoptosis and cell cycle arrest.

There is growing interest in developing potent modulators to impede NF- κ B action by inhibition of different steps of the NF- κ B signaling pathway. The repression of up-stream proteins of NF- κ B activation, that lead to NF- κ B inhibition is the vital step to down the tumor growth and drug resistance (Zeligs et al. 2016). A significant advance in our present study is to understand the ability of combinatorial curcumin and tangeretin on up-stream proteins of NF κ B pathway that dispense the novel therapeutic strategies targeting this signaling pathway.

2 Methodology

2.1 Binding pocket identification

The binding pocket analysis of IKK complex, Proteosome, Rel A (p⁶⁵) and NF- κ B proteins was performed using Computed

Atlas of Surface Topography of Proteins (CASTp) program (<http://cast.engr.uic.edu>). CASTp evaluates the active site, pocket area and binding pocket volume of proteins.

2.2 Protein preparation

The crystal structures of IKK Complex (PDB ID: 3BRV), Proteosome (PDB ID: 1D3Z), p⁶⁵ (PDB ID: 1NFI) and NF- κ B (PDB ID: 3GUT) were retrieved from the Protein Data Bank (<http://www.rcsb.org/>). Prior to initiating the docking simulations, all non-protein molecules were removed, for any alternative atoms locations, only the required location was retained. All target proteins were energy minimized before docking studies.

2.3 Ligand preparation

The 3D structures of curcumin ((1E, 6E)-1, 7-bis(4-hydroxy-3-methoxyphenyl) hepta-1, 6-diene-3, 5-dione) (mw: 368.3799 g/mole) and tangeretin (5, 6, 7, 8-tetramethoxy-2-(4-methoxyphenyl) chromen-4-one) (mw: 372.3686 g/mol) were retrieved from the Pubchem database (<http://pubchem.ncbi.nlm.nih.gov/>) and pre-optimized by applying MMFF94x force field (Fig. 1a, b). The optimized structures were used as initial conformations for the molecular docking studies.

2.4 Molecular docking

AutoDock v4.0 All docking simulations were carried out by AutoDock v4.0 using the Lamarckian genetic algorithm

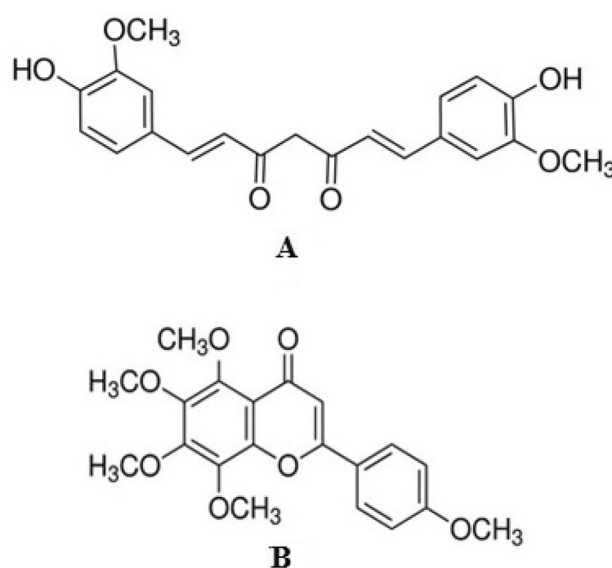


Fig. 1 Chemical structures of **a** curcumin and **b** tangeretin

(Huey et al. 2007). In docking, the target proteins were kept rigid, therefore assuming there is no induced conformational change upon ligand binding. In contrast to the protein, torsional flexibility was permitted for the ligands via the side-groups and backbone of protein was kept rigid. Atomic affinity and electrostatic potentials were computed for a grid box positioned around the approximate center of the binding site, with dimensions around $40\text{\AA} \times 40\text{\AA} \times 40\text{\AA}$ using grid point spacing of $0.375\text{\AA}$ (roughly a quarter of the length of a carbon–carbon single bond). The grid maps are calculated by AutoGrid. The Lamarckian genetic algorithm (LGA) is a docking protocol which is built in algorithms of Auto Dock. In LGA, population size was increased to 300 and the number of energy evaluations per trial was set to 5×10^6 . Genetic algorithms usually identify the notion of an individual and its genetic code with respect to the problem's solution space. Auto dock treats a particular orientation of the ligand inside the protein as an 'individual' in its GA. The increase was due to the large number of flexible torsions that are explored during simulation, which increase the search space for the more flexible compounds. Each flexible docking simulation was repeated for 100 trials, yielding 100 docked conformations for each ligand. AutoDock v4.0 calculates the experimental free energy functions for estimating binding energy, inhibitory constant, intermolecular energy, internal energy and torsional energy. The binding free energy was empirically calculated based on the energy terms and a set of co-efficient factors (Morris et al. 1999). The binding energy was used to represent the hierarchy of docking positions. The more clusters with lowest value of binding energy were selected as the best docking pose for ligand. The docked ligand-receptor complexes were analyzed by PyMOL (Delano 2006).

HEX docking Hex is an interactive molecular graphics program for calculating and displaying feasible docking modes of pairs of proteins and DNA molecules. Hex can also calculate protein–protein docking, protein–ligand docking, assuming the ligand is rigid, and it can superpose pairs of molecules using only knowledge of their 3D shapes.

In the present study, we used hex docking server for combined docking studies. Pre-prepared AutoDock files

of ligand and receptors are used as pdb's for hex docking. All parameters were kept as default and performed docking simulations. The obtained ligand-receptor complexes were further analyzed by PyMOL visualizing tool (<http://www.pymol.org>) (Delano 2006).

3 Results

3.1 Active site identification

Active site identification was performed by CASTp program (Binkowski et al. 2003). The predicted parameters such as pocket volume, pocket ID, surface area, and active site residues for each protein are represented in Table 1 and proteins with ligand binding pockets are displayed in Fig. 2.

3.2 Binding interactions of curcumin and tangeretin with NF- κ B cascade proteins

The molecular interactions of NF- κ B cascade proteins and ligands were performed by AutoDock v4.0 and hex docking tools. Stack of all the conformations of lead molecules in docking space was observed. Clusters of lead molecules were generated about 100 conformations in each run of AutoDock v4.0. Free energy of binding, inhibitory constant (K_i), and RMSD are important parameters considered in the screening of ligand-receptor interactions. The combined docking studies were carried out using hex server.

3.3 Docking study of NF- κ B cascade proteins using AutoDock v4.0

3.3.1 Docking study of IKK complex

The AutoDock analysis was performed for crystal structure of IKK complex against ligands (curcumin and tangeretin). The five dominant clusters ranked with the lowest energy scores and ligand conformations of best cluster were

Table 1 Ligand binding pocket, surface area, volume and active site residues of NF- κ B cascade proteins obtained by CASTp program

Protein	Pocket ID	Pocket area ($\text{\AA}^2$)	Pocket volume ($\text{\AA}^3$)	Amino acids at binding pocket
IKK Complex	28 and 29	92 and 338.1	92.5 and 391.1	Met ⁹⁴ , Phe ⁹⁷ , Gln ⁹⁸ , Arg ¹⁰¹ , Leu ⁷³⁷ , Asp ⁷³⁵ , Trp ⁷³⁹ , Trp ⁷⁴¹ , Ser ⁷³³
Proteasome	10	51.5	48	Glu ²⁴ , Lys ²⁷ , Pro ³⁸ , Asp ⁵² , Asp ³⁹
P ⁶⁵ (Rel A)	44	52.2	72.2	Arg ³⁰² , Thr ³⁰⁵ , Tyr ³⁰⁶ , Phe ³⁰⁹
NF- κ B	108	1699.7	4979.7	Arg ³³ , Tyr ³⁶ , Lys ³⁷ , Cys ³⁸ , Glu ³⁹ , Arg ¹²⁴ , Lys ¹²² , Asp ¹²⁵ , Glu ¹²⁸ , Ala ¹²⁹ , Val ¹²¹ , Gln ¹³² , Asn ¹⁸⁶ , Lys ²¹⁸ , Lys ²¹⁸ , Arg ²⁴⁶ , Tyr ³⁵⁷ , Val ³⁵⁸ , Glu ³⁶⁰ , His ⁴⁴¹ , Val ⁴⁴² , Lys ¹²³ , Thr ⁴⁴³ , Lys ⁴⁴⁴ , Lys ⁵⁰³ , Glu ⁵⁰⁴ , Ser ⁵⁰⁸ , Lys ⁵⁴¹ , Arg ⁶⁰⁵ , Gln ⁶⁰⁶ , Ala ⁵⁴²

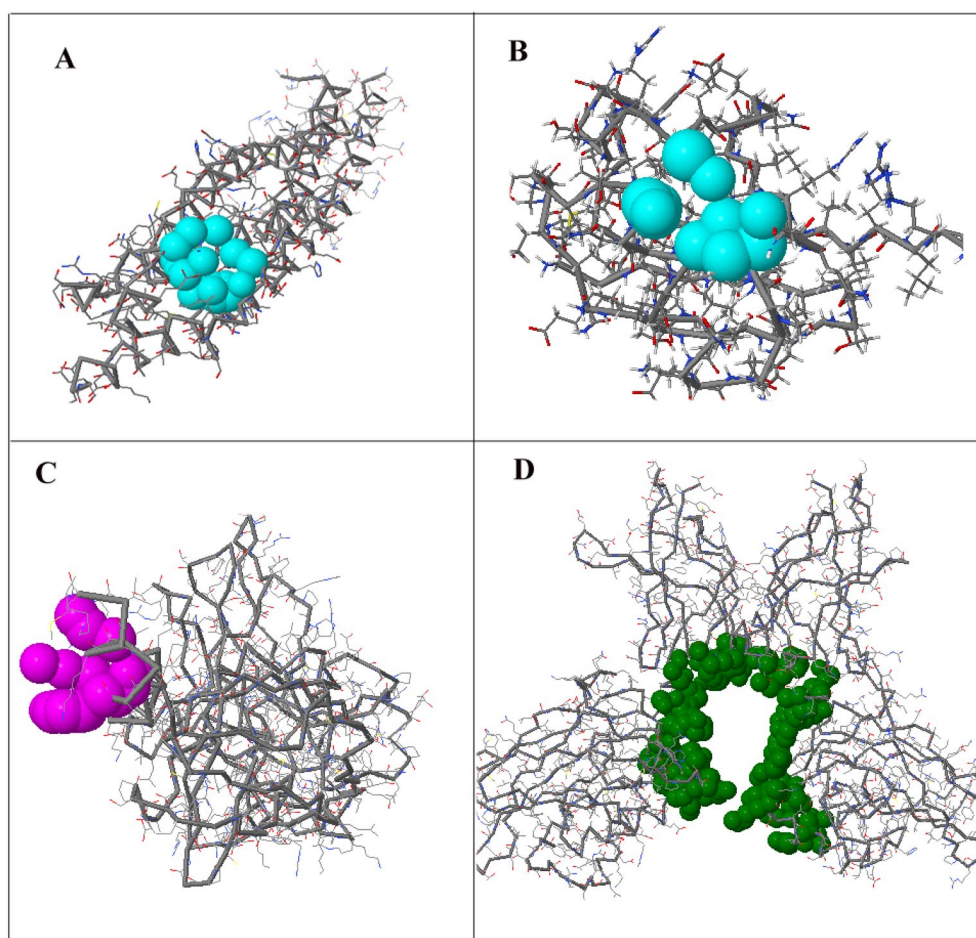


Fig. 2 Active site pockets of **a** IKK complex, **b** proteasome, **c** Rel A and **d** NF- κ B protein—predicted using CASTp program

selected. The docked conformations of curcumin and tangeretin showed interactions with IKK complex active site residues of Leu⁷³⁷, Asp⁷³⁸, Trp⁷³⁹ and Lys⁹⁶ (Fig. 3a) and Thr⁷³⁵, Ser⁷³³ (Fig. 4a), respectively. The docking energy of curcumin in cluster 40 is -4.13 kcal/mol and tangeretin in cluster 35 is -2.76 kcal/mol (Table 2).

3.3.2 Docking study of proteasome

The docking results of curcumin and tangeretin against proteasome are shown in Table 2. As noted before good convergence was achieved, with the best-docked conformations also those found to have the lowest binding energy and the greatest number of conformations per cluster. Conformations appear a strong hydrogen bond interaction of curcumin and tangeretin with proteasome active residues Gln⁴⁹, Arg⁴², Asp⁵², Glu²⁴ (Fig. 3b) and Asp⁵² (Fig. 4b), respectively. Concerning the docking results, the docked energy in cluster 7 was -4.38 kcal/mol (curcumin) and cluster 22 was -4.05 kcal/mol (tangeretin).

3.3.3 Docking study of Rel A (p65)

Docking analysis of curcumin and tangeretin on Rel A (p65) protein has shown binding energy in cluster 32 is -6.00 kcal/mol and cluster 35 is -3.46 kcal/mol, respectively (Table 2). The docked conformations showed an interaction of curcumin and tangeretin with Rel A protein residues; Arg³⁰², Arg³⁰⁴ and Thr³⁰⁸ (Fig. 3c) and Arg³⁰² and Thr³⁰⁵ (Fig. 4c), respectively.

3.3.4 Docking study of NF- κ B complex

In docking conformations of NF- κ B; curcumin and tangeretin have been reported that they interact with binding residues of Cys³⁵⁹, Lys⁴⁴⁵, Lys⁵⁴¹ and Glu³⁶⁰ (Fig. 3d), and Lys⁴⁴⁴ and Tyr³⁵⁷ (Fig. 4d), respectively. Concerning the docking results, the docked energy in cluster 29 is -5.42 kcal/mol (curcumin) and the docked energy in cluster 2 is -5.31 kcal/mol (tangeretin). The docking results are represented in Table 2.

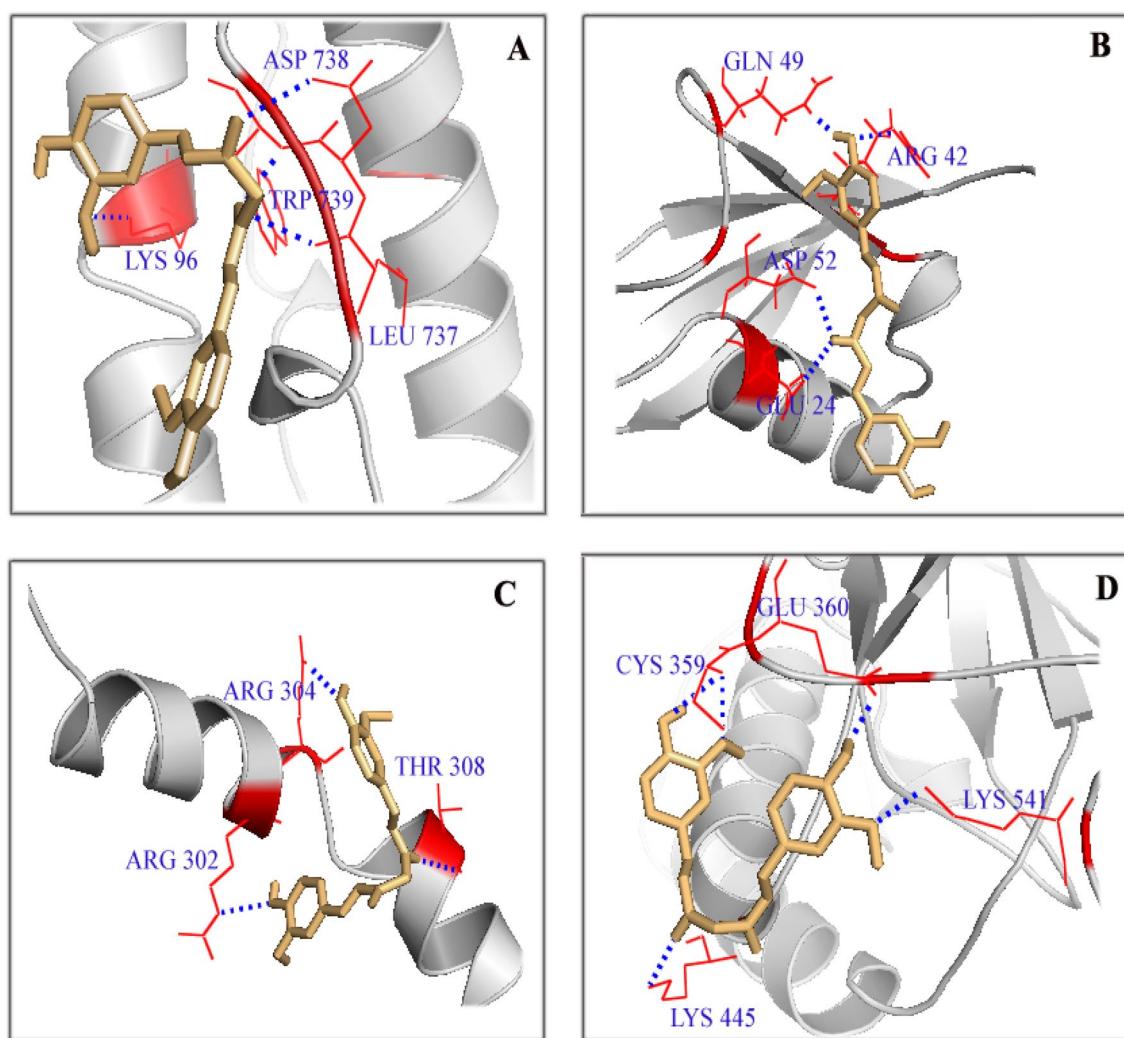


Fig. 3 Binding interaction of curcumin with **a** IKK complex, **b** 26S proteasome, **c** p65 (NLS) and **d** NF-kB. (Blue dotted lines represent H-bonds and light orange sticks represent curcumin) (color figure online)

3.4 Hex docking

In combined docking approach, the ligand conformations of curcumin and tangeretin on active sites of NF-kB signaling proteins have shown best docking interactions, compared to the results obtained in AutoDock. Concerning the hex docking results, the docking energies of IKK complex (−88.28 kcal/mol), proteasome (−98.64 kcal/mol), Rel A (−53.89 kcal/mol) and NF-kB complex (−80.55 kcal/mol) are represented in Table 2. The hex docking confirmations of NF-kB signaling proteins with both ligands are displayed in Fig. 5a–d.

Based on hex docking results, proteasome has shown highest docking energy of −98.64 kcal/mol in combination of curcumin and tangeretin. In combined form, the ligands have shown more potential to bind with NF-kB cascade proteins compared to individual docking study.

4 Discussion

The plant-derived natural compounds can check and prevent the onset of cancer progression and other inflammatory diseases. The natural compounds have been used as medicine for human diseases over past several decades due to their non-toxic nature and cause fewer side effects when compared to chemotherapeutic drugs. Furthermore, combination therapy is often considered to be the best treatment for prevention of onset of several diseases. The combined form of natural compounds or used as adjuvants along with chemotherapeutic drugs could enhance the therapeutic index and ability to modulate multiple signaling transduction pathways involved in cancer and inflammations. Curcumin is the most powerful plant compound possesses antioxidant, anti-inflammatory, and anticancer properties; and it has been used as adjuvant in combination with other phytochemicals

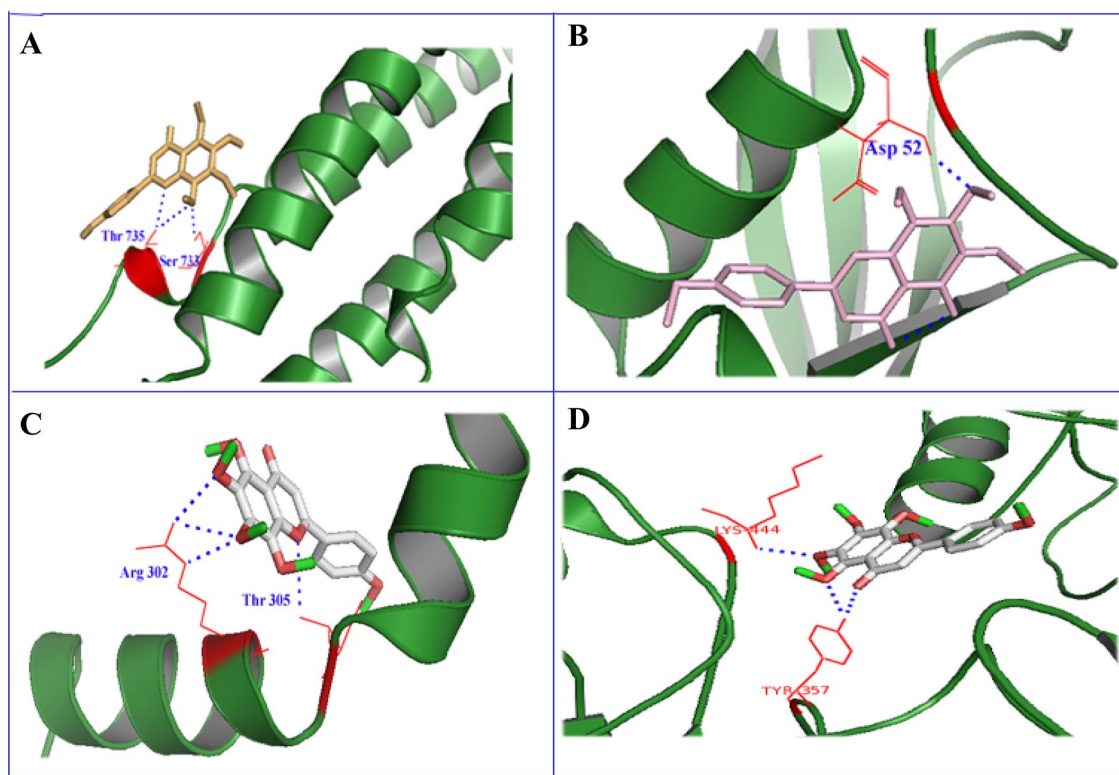


Fig. 4 Binding interaction of tangeretin with **a** IKK complex, **b** 26S proteasome, **c** p65 (NLS) and **d** NF- κ B. (Blue dotted lines represent H-bonds and stick conformations represent tangeretin) (color figure online)

Table 2 Auto Dock and Hex docking energies (kcal/mol) of NF- κ B pathway proteins and their active site residues interaction with curcumin and tangeretin

Protein	Ligand	Lowest binding energy (Kcal/mol)			Amino acids involved in H-bonding
		Auto dock	Hex	Cur-cumin + tangeretin	
IKK complex	Curcumin	-4.13	-118.419	-88.28	Leu ⁷³⁷ , Asp ⁷³⁸
	Tangeretin	-2.76	-98.687		Trp ⁷³⁹ , Lys ⁹⁶ , Thr ⁷³⁵ , Ser ⁷³³
Proteasome	Curcumin	-4.38	-89.721	-98.64	Gln ⁴⁹ , Gln ⁴¹
	Tangeretin	-4.05	-77.003		Arg ⁴² , Glu ²⁴ , Asp ⁵² , Asp ⁵²
Rel A (p65)	Curcumin	-6.00	-57.448	-61.09	Arg ³⁰⁴ , Thr ³⁰⁸ , Arg ³⁰²
	Tangeretin	-3.46	-49.402		Arg ³⁰² , Thr ³⁰⁵
NF- κ B	Curcumin	-5.42	-93.209	-80.55	Lys ⁴⁴⁵ , Lys ⁵⁴¹
	Tangeretin	-5.31	-78.044		Glu ³⁶⁰ , Cys ³⁵⁹ , Lys ⁴⁴⁴ , Tyr ³⁵⁷

such as quercetin, retinoic acid and resveratrol in cancer treatment (Sinha et al. 2012). Tangeretin, a natural polymethoxyflavone exhibits antiproliferative and antioxidant activities (Rooprai et al. 2001). In vivo studies reported that tangeretin suppressed the inflammatory response by inhibition of TNF- α , IL-12, and IL-23 expression and NF- κ B activation in the colon of mice with colitis (Eun et al. 2017). The combination treatment of tangeretin and cisplatin in cancer cells enhanced the antitumor effects by down-regulation of NF- κ B and PI3K/Akt signal transduction pathways (Arafa-el

et al. 2009). The methods to inhibit NF- κ B activation in cancers are considered a worthwhile addition to the current therapeutic strategy. In the present study, first time we performed the combined docking analysis of curcumin and tangeretin on NF- κ B signaling proteins.

The IKK complex is a key component in NF- κ B signaling pathway that activates I κ B α through phosphorylation of Ser32 and Ser36 (Rusha et al. 2008). In cancer cells, over expression of IKK kinase induces NF- κ B hyper activation. The IKK complex is composed of catalytic IKK α / β kinases

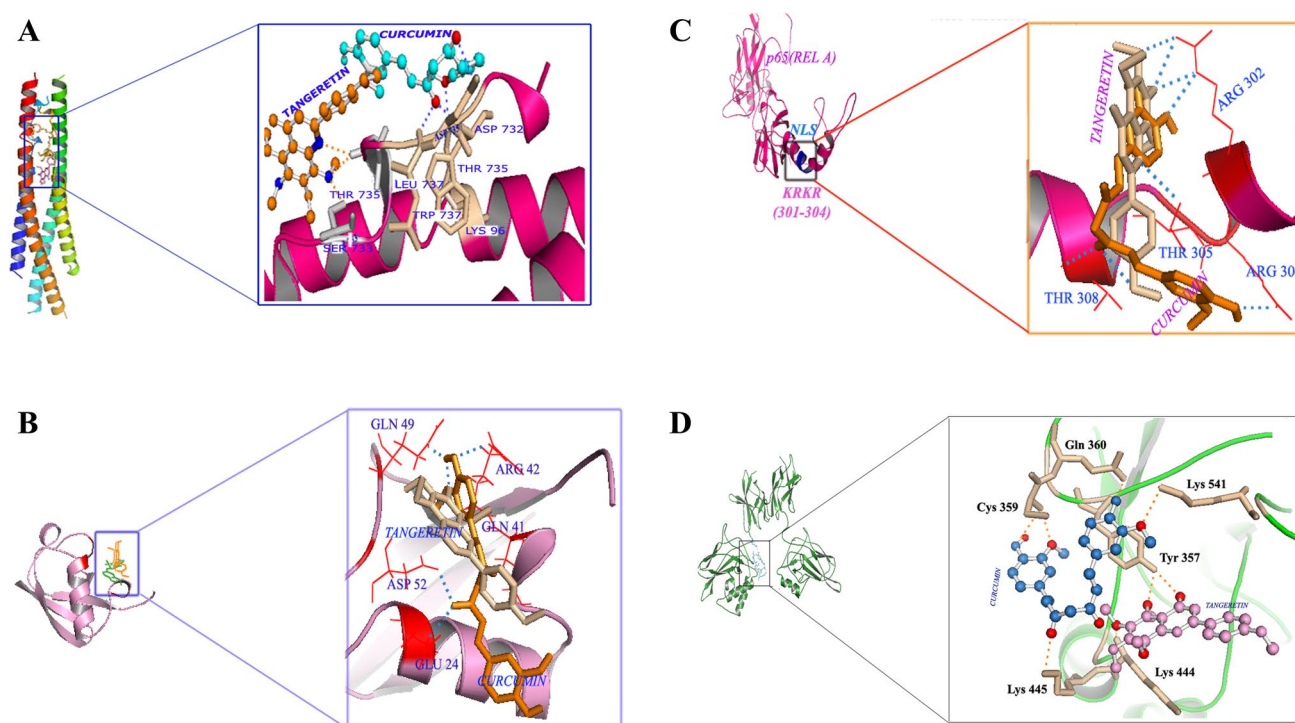


Fig. 5 Hex docking conformations of combination of curcumin and tangeretin with **a** IKK complex, **b** 26S proteasome, **c** p65 (NLS) and **d** NF-kB complex

and a regulatory protein (NEMO or IKK γ - NFkB essential modulator). The catalytic activity of IKK complex is activated by the association of NEMO with the un-phosphorylated IKK kinase C termini. The protein–protein interactions of IKK and NEMO were contributed by the amino acid residues from IKK kinase (W⁷⁴¹, W⁷³⁹ and F⁷³⁴) and NEMO (Leu⁹³, Phe⁹⁷, Lys⁹⁰, Glu⁸⁹ and Val¹⁰⁴) (Rusha et al. 2008). In the present study, we reported that both curcumin and tangeretin were interfering the association of IKK-NEMO complex by interacting with active site residues (Leu⁷³⁷, Asp⁷³⁸, Trp⁷³⁹, Thr⁷³⁵ and Ser⁷³³) thereby preventing the IKK complex formation. Previous reports have been suggested that curcumin inhibits NF-kB activation mediated gene expression through inhibition of IKK complex in cancer cells (Kim et al. 2011, 2012; Marquardt et al. 2015). However, the molecular mechanism of curcumin interaction with IKK kinase was still unclear. In this study, we evaluated the binding nature of curcumin along with other phytochemical, tangeretin. The combined form of curcumin and tangeretin destabilize the IKK-NEMO complex formation thereby, may inhibit the NF-kB activation. The detailed molecular interactions of curcumin and tangeretin are represented in Figs. 3a and 4a, respectively and the combined form of ligand interactions are displayed Fig. 5a.

Cancer cells have highly proliferative nature and can divide more rapidly than normal cells. The rate of protein synthesis and degradation is higher in cancer cells when

compared to normal cells. 26S proteasome (a multienzyme protease) mediates the degradation of cellular unnecessary proteins; conversely, blockade of this function in cancer cells results in cellular death. Several proteasome inhibitors are already developed, but they are showing side effects in cancer patients. For example, Bortezomib is the first proteasome inhibitor approved by the food and drug administration (FDA) has been used for the treatment of multiple myeloma (Kane et al. 2007). Proteasome degrades the I κ B kinase in NF-kB activation pathway. Various research groups were reported that curcumin act as proteasome inhibitor, that can inactivate the NF-kB and increased p53 levels in cancer cells (Jana et al. 2004; Banerjee et al. 2018). In cancer mouse model and cell lines, curcumin disturbs 26S proteasome activity via dual-specificity tyrosine-regulated kinase 2 (DYRK2) inhibition leading to impairment of cell proliferation (Banerjee et al. 2018). Combination of curcumin and garcinol (a bioactive compound derived from the rind of the fruit, *Garcinia indica*) exhibit a high level of synergism on human pancreatic cancer cells and increased apoptosis was observed compared with the effect of individual agents (Parasramka et al. 2012). Curcumin combination with berberine (Wang et al. 2016) and citral (Patel et al. 2015) exhibit synergistic effect on breast cancer cells through distinct molecular mechanisms. Co-combination of curcumin and resveratrol exhibits chondroprotective effects in human articular

chondrocytes through inhibition of NF- κ B mediated gene expression. These compounds suppress the I κ K and proteasome activation, inhibition of I κ B α phosphorylation and degradation, and inhibition of nuclear translocation of NF- κ B (Csaki et al. 2009). Despite its mode of action, the molecular mechanism of curcumin and tangeretin mediated proteasome inhibition was not reported earlier. The test molecules examined in the present study—curcumin (Fig. 3b) and tangeretin (Fig. 4b) have shown binding interactions with active site residues of proteasome (Gln⁴⁹, Arg⁴², Asp⁵², Glu²⁴) thereby imbibing its action. The combined form of curcumin and tangeretin mediated proteasome inhibition is represented in Fig. 5b.

Rel A, a subunit of NF- κ B complex consisting of N-terminal signal sequence (KRKR), which plays a key role in the translocation of NF- κ B into the nucleus. In the inactive state of NF- κ B, I κ B masks the NLS at Rel A thereby retain the NF- κ B in cytosol. In NF- κ B activation process, the I κ B is degraded by proteasome which leads to the consequent release of NLS from the I κ B inhibition. The specific interaction of curcumin (Fig. 3c) and tangeretin (Fig. 4c) with key residues of NLS (Arg³⁰², Arg³⁰⁴ and Thr³⁰⁸), which mimics the action of I κ B. The combined molecular interactions of curcumin and tangeretin are displayed in Fig. 5c.

The DNA binding pocket of NF- κ B complex consists of active site residues, which are mainly, interact with DNA during NF- κ B dependent gene expression. The reduction of binding interaction with DNA may leads to inhibition of NF- κ B dependent of gene expression. Tangeretin suppresses the overexpression of pathways like PI3K/Akt, MAPK and NFB signaling in prostate cancer (Guo et al. 2015). Several research groups reported that curcumin is an excellent inhibitor of NF- κ B in various cancers and inflammatory diseases (Kim et al. 2012; Iqbal et al. 2018). The combined action of curcumin and tangeretin on NF- κ B inhibition was not reported earlier. In this study, we performed bioinformatics analysis of curcumin and tangeretin binding mechanisms with NF- κ B. These compounds showed best fit orientation at the active site of NF- κ B (Cys³⁵⁹, Lys⁴⁴⁵, Lys⁵⁴¹, Lys⁴⁴⁴, Tyr³⁵⁷ and Glu³⁶⁰) (Figs. 3d and 4d). Thus, binding of curcumin and tangeretin at active site of NF- κ B could change its original conformation and prevents interaction with concerned gene (Fig. 5d).

We strongly believe that our computational results will direct the future experimental studies on curcumin and tangeretin combination for anticancer activity in in vitro and in vivo models.

5 Conclusions

In drug discovery, the combined form of approaches may an essential and potential route for targeting the multiple diseases including drug-resistant microbes. The combined

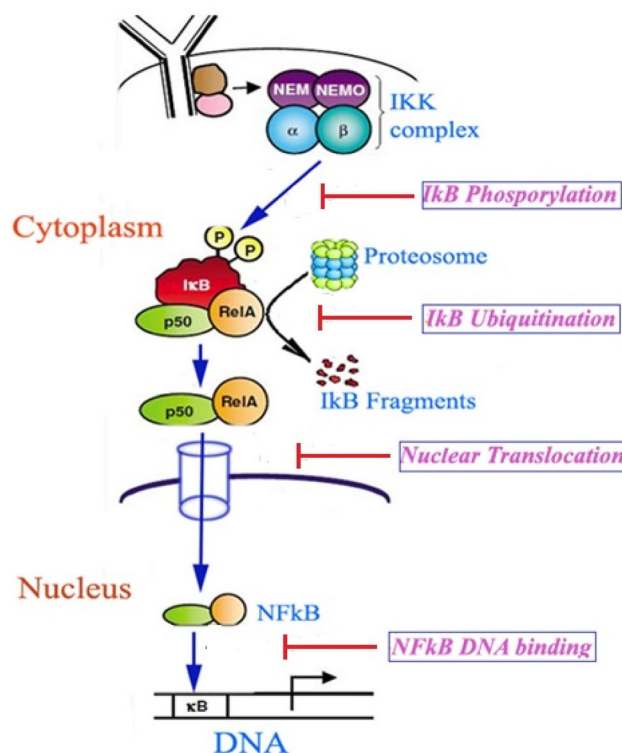


Fig. 6 Schematic illustration of drug targets and inhibition site of curcumin and tangeretin in NF- κ B signaling pathway

form of drugs which control the complex disease systems are less prone to drug resistance and are the standard of care in many important therapeutic areas. The combinations of drugs are currently employed in rational design, but the increased efficacy provides justifies an in vitro discovery efforts for identifying novel multi-target mechanisms. A drug combination represents a systematic approach to identify interactions between molecular pathways that could be leveraged for therapeutic benefit.

In conclusion, the natural compounds curcumin and tangeretin have shown a strong binding affinity and docking interactions with NF- κ B cascade proteins. The inhibition sites for curcumin and tangeretin in NF- κ B pathway are represented in Fig. 6. In individual docking (AutoDock) with each protein, the ligands have shown various degrees of interactions and docking energies. In the combined docking approach (Hex docking), the proteasome has shown highest docking energy compared to other target proteins. The present study is concluded that the combined form of compounds may effectively control the NF- κ B activation through proteasome inhibition. With synergic mode of action, the combined approach is a good way to inhibit the protein targets in multiple diseases.

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